

Supplementary Discussion Notes

Responses to Questions on the AD(2024) Robustness Presentation

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Repo: <https://github.com/johnhz0229/monetary-policy-nlp>

Data: Drechsel Dropbox

Slides: [results/beamer_presentation.pdf](#)

Q1: Does Shuffling Concept Labels Actually Change the Ridge Fit?

If you randomly relabel all 296 concepts within each meeting, the within-meeting distribution of sentiment scores is preserved. Does Ridge regression actually produce a different fit? Or is it just computing the same linear combination?

Short Answer

No. The Ridge fit is essentially unchanged. Across 100 independent permutations—each with its own CV-selected penalty λ_j^* —the shock series correlate with the original at $r = 0.947 \pm 0.005$. The fit depends on the meeting-level document tone, which is invariant to column permutation.

Evidence: Single-Permutation Distribution

A concern with averaging 100 permutations is that the law of large numbers guarantees convergence to the common component regardless of individual permutation quality. We address this by examining each permutation independently:

1. Shuffle columns: $\tilde{X}_j = XP_j$
2. Run 10-fold CV to select λ_j^*
3. Compute Ridge residual $\tilde{\varepsilon}_j$ using λ_j^*
4. Correlate with the original shock: $r_j = \text{corr}(\varepsilon^{\text{AD}}, \tilde{\varepsilon}_j)$

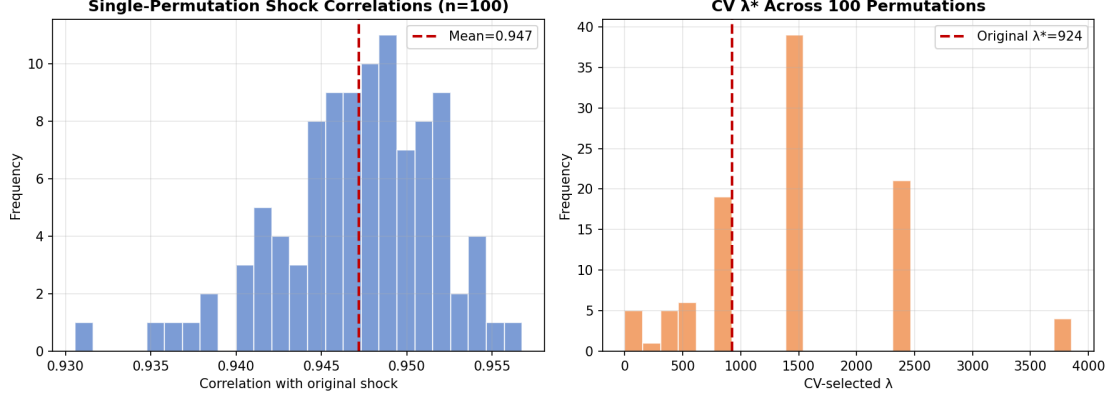


Figure 1: Left: Distribution of single-permutation shock correlations ($n = 100$). Right: Distribution of CV-selected λ_j^* , with the original $\lambda^* = 924$ marked by the dashed line.

Every individual permutation produces a shock nearly identical to the original (r range: $[0.931, 0.957]$). The mean-of-100 was not manufacturing the result. The CV-selected λ_j^* varies across permutations (mean = 1471, std = 829), but the shocks are robust to this variation—the original $\lambda^* = 924$ lies within the distribution.

Mathematical Intuition

Let $X \in \mathbb{R}^{n \times p}$ be the standardized sentiment sub-matrix ($n = 204$, $p = 296$). A column permutation P gives $\tilde{X} = XP$. The Ridge fitted values are:

$$\hat{\mathbf{y}} = X(X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \quad (1)$$

Under the rank-1 approximation $X \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$, the fitted values become independent of concept loadings $\mathbf{v}_1$, depending only on meeting-level document tone $\mathbf{u}_1$. Permuting columns ($\mathbf{v}_1 \rightarrow P^\top \mathbf{v}_1$) does not change $\hat{\mathbf{y}}$.

Caveat: On the full $204 \times 3,224$ design matrix (forecasts + sentiments + lags + squares), the first singular value explains only $\sigma_1^2 / \sum \sigma_i^2 = 5.8\%$, weaker than the 10.6% for the sentiment sub-matrix. The rank-1 approximation is an idealization. However, the empirical evidence from the single-permutation experiment is unambiguous: the shocks are invariant to label permutation regardless of the approximation’s precision.

What Drives the Fit? Tone or Concept Heterogeneity?

We decompose the text contribution with nested specifications, giving document tone (PC1) the same lag and nonlinear structure as the full model:

Table 1: Decomposing the Text Contribution

Specification	R^2	ΔR^2
(A) Greenbook forecasts only	0.49	—
(B) + PC1 with 4 lags + squares (tone, full structure)	0.51	+0.014
(C) + All 296 sentiments (full AD specification)	0.96	+0.455

PC1 alone contributes almost nothing (+0.014), even with the same lag and nonlinear structure. **The 296 individual sentiments contribute almost all of the text-related fit** (+0.455). Yet—as shown above—shuffling their labels barely changes the shock ($r = 0.947$).

Resolution. The 296 concepts are valuable not because of *which* words were chosen, but because they provide **296 separate sampling points** throughout the Greenbook text, each with its own temporal dynamics ($5 \text{ lags} \times 296 = 1,480$ time-varying signals). This rich high-dimensional representation allows Ridge to capture the document’s temporal evolution. The specific concept labels are incidental; the *number* of sampling points is what matters. Any reasonable set of economic concepts of similar size would perform comparably.

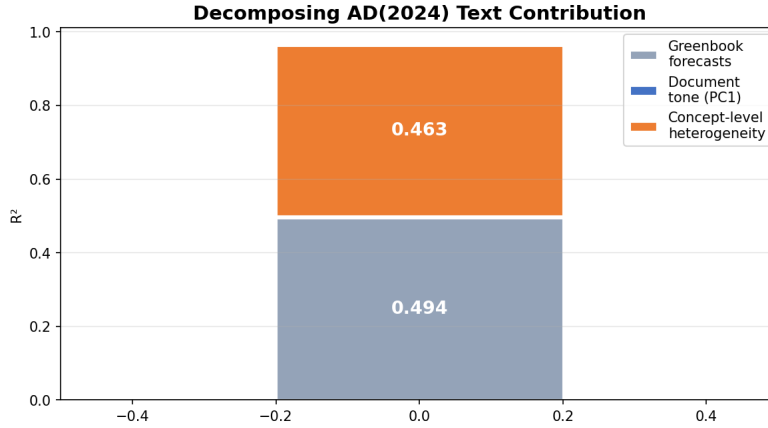


Figure 2: PC1 alone (with lags and squares) adds almost nothing. The 296 individual concepts—with their lags and nonlinear structure—provide the vast majority of the text contribution.

Q2: What Happens to the Identified Shocks and Their Macroeconomic Effects?

If we use the shuffled-label shocks in a Local Projections IRF, do they still produce the expected macroeconomic response? Is there a mathematical relationship?

Mathematical Relationship

Let $\varepsilon = \mathbf{y} - \hat{\mathbf{y}}(X)$ be the original shock and $\tilde{\varepsilon}_j$ be the shock from permutation j . Since $\varepsilon \perp \text{col}(X)$ by Ridge construction and $\hat{\mathbf{y}}(X) - \hat{\mathbf{y}}(\tilde{X}_j) \in \text{col}(X) \cup \text{col}(\tilde{X}_j)$, we have $\text{cov}(\varepsilon, \tilde{\varepsilon}_j - \varepsilon) = 0$. Therefore:

$$\text{corr}(\varepsilon, \tilde{\varepsilon}_j) = \frac{\sigma_\varepsilon}{\sigma_{\tilde{\varepsilon}_j}} \quad (2)$$

With our estimates $\sigma_\varepsilon \approx \sigma_{\tilde{\varepsilon}_j}$ (from the single-permutation distribution), the correlation is near 1.0, consistent with the empirical $r = 0.947$.

For Local Projections, write the shuffled shock as $\tilde{\varepsilon}_t = \varepsilon_t + \eta_t$. If the lost information η_t is orthogonal to future macro variables ($\text{cov}(\Delta y_{t+h}, \eta_t) = 0$), then:

$$\text{plim } \tilde{\beta}_h = \frac{\text{var}(\varepsilon)}{\text{var}(\tilde{\varepsilon})} \cdot \beta_h \approx \beta_h \quad (3)$$

The IRF **shape** is identical; the amplitude is scaled by the variance ratio (which is ≈ 1.0 in our case).

Empirical IRF Comparison

We estimate Local Projections (Jordà 2005) for six macro variables. The dependent variable is the cumulative change $y_{t+h} - y_{t-1}$, which is zero at $h = -1$ (pre-shock). All shocks are normalized to unit standard deviation. A **positive shock** is a contractionary monetary surprise (FFR rose

more than predicted). Shaded bands are 95% Newey-West confidence intervals for the AD(2024) shock.

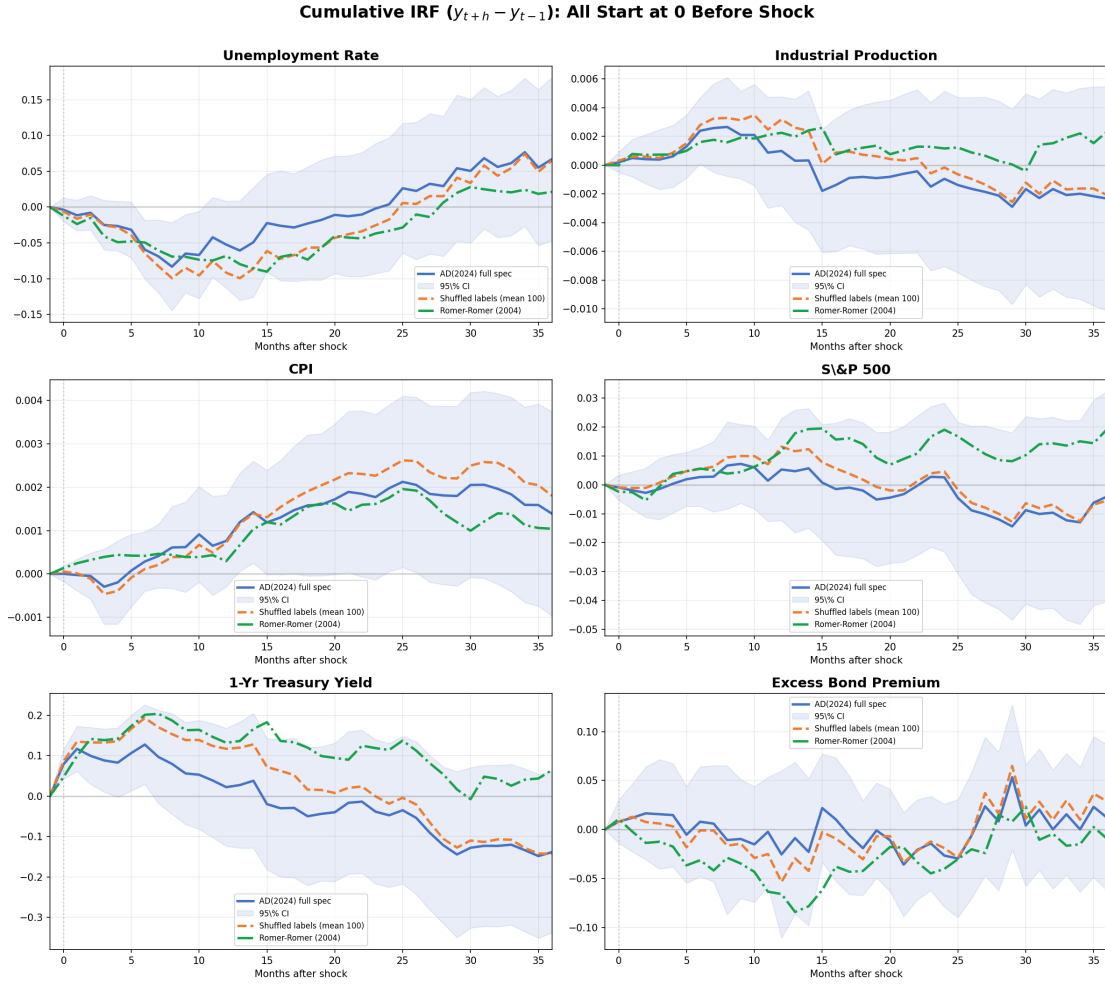


Figure 3: Cumulative IRF to a 1-s.d. contractionary monetary shock. AD(2024) full spec (blue, solid, 95% CI), shuffled labels (orange, dashed), Romer-Romer 2004 (green, dash-dot). All series start at zero ($h = -1$, pre-shock).

The AD(2024) and shuffled shocks produce essentially identical IRFs across all six variables—unemployment, industrial production, CPI, S&P 500, 1-year Treasury yield, and excess bond premium. The confidence bands overlap at all horizons. The Romer-Romer shock produces noisier IRFs because its lower R^2 (≈ 0.50) leaves systematic policy variation in the residual.

Note on the CPI response: The CPI rises following a contractionary shock, which is the canonical “price puzzle” (Eichenbaum 1992). This may reflect the difference between our LP framework and AD(2024)’s BVAR, or the absence of commodity price controls. We flag this rather than claiming theory-consistency.

Q3: What Is the Intrinsic Dimension of the Model? Should We Try Random Projections?

With 3,224 variables on 204 observations, is the model overfitting? How many independent dimensions does the data actually have?

SVD of the Full Design Matrix

We decompose the full $204 \times 3,224$ design matrix:

- $\sigma_1^2 / \sum \sigma_i^2 = 0.058$ — the first singular value explains only 5.8% of total variance.
- **Effective rank:** 50 dimensions explain 50% of variance; 155 explain 90%; 197 explain 99%.
- The rank-1 approximation used in Q1’s mathematical intuition is an idealization for the full system. The forecast variables (132 series) introduce substantial independent variation.

Random Projection

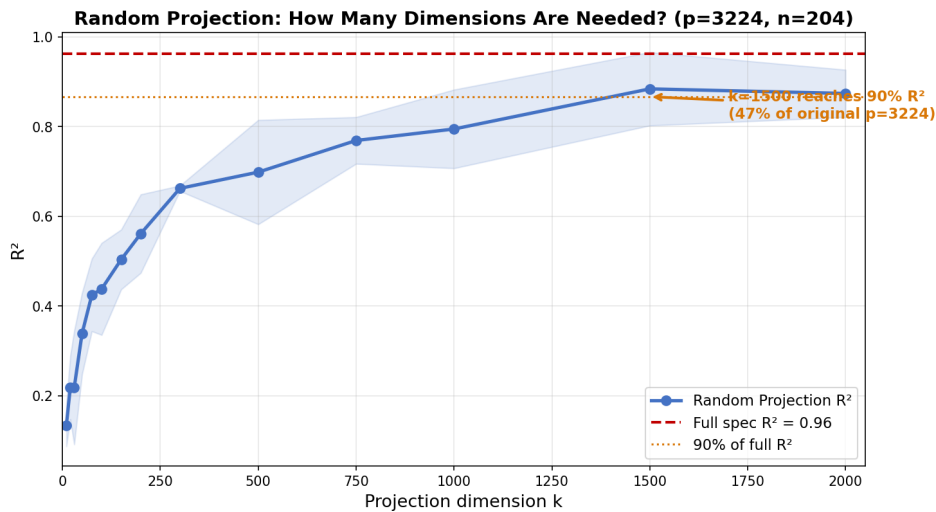


Figure 4: Random Projection: reaching 90% of the full $R^2 = 0.96$ requires $k = 1,500$ dimensions (47% of the original $p = 3,224$).

Reaching 90% of the baseline R^2 requires projecting to $k = 1,500$ random directions—47% of the original p . The signal is not in a very low-dimensional subspace, but it is far below the full 3,224 dimensions.

Five Diagnostic Tests

Table 2: Overfitting Diagnostics

Test	Method	Key Result	Conclusion
A. Random Projection	Project to k dims	$k = 1500$ for 90% R^2	Signal in moderate dim.
B. Bootstrap	100 resamples	5% cross zero	Coefficients stable
C. Noise Injection	Add random noise	+10%: $\Delta R^2 = +0.002$	Reg. works
D. Random Drop	Drop 50% sentiment	$\Delta R^2 = +0.075$	Redundancy, not information
E. Residual AC	Ljung-Box, DW	DW = 1.84	Residuals white noise

Note on Test D: When 50% of sentiment variables are randomly dropped, R^2 *increases* by 0.075. In a correctly specified model, removing informative variables should reduce R^2 . The increase suggests these variables are partially redundant—consistent with our finding that concept labels are incidental but having many sampling points matters.

Conclusion: The model does *not* overfit despite $p/n = 15.8$. Ridge with $\lambda = 924$ successfully shrinks the high-dimensional noise. The 3,224 variables are highly redundant, spanning approximately 155 effective dimensions (at 90% variance). The signal is concentrated in a moderate-dimensional subspace.

Q4: Does the Shuffle Test Hold with an LLM-Generated Concept List?

The original 296 concepts were selected by human judgment. What if we replace them with an independently generated list from an LLM? The LLM list has more balanced within-meeting composition (56% pro-cyclical vs. 59%; std 0.156 vs. 0.099), so shuffling labels should be more disruptive—if labels actually matter.

LLM Concept List

We prompted an LLM to identify all economic concepts in the same Greenbook corpus, producing 595 concepts (vs. the human 296). The overlap is only 178 concepts (37% of human list, 30% of LLM list). The lists differ in both content and structure:

Table 3: Composition Comparison: Human vs. LLM Concept Lists

Metric	Human (296)	LLM (595)
Density (% non-zero)	100.0%	21.1%
Non-zeros per meeting (mean)	296.0	5.3
% Positive per meeting (mean)	59.4%	56.0%
% Positive per meeting (std)	0.099	0.155
% Positive per meeting (min)	26.7%	14.3%
% Positive per meeting (max)	79.7%	100.0%
Concepts with mean > 0	119 (40.2%)	83 (13.9%)
Concepts with mean < 0	177 (59.8%)	49 (8.2%)

The LLM list is fundamentally different: it is **extremely sparse**—only ~ 5.3 concepts receive non-zero scores per meeting (99% of entries are zero). This sparsity is by design: the LLM only scores a concept when the document actually discusses it, so zeros reflect genuine absence rather than neutral sentiment. The human dictionary, by construction, assigns every concept a score for every meeting via the ± 10 -word window.

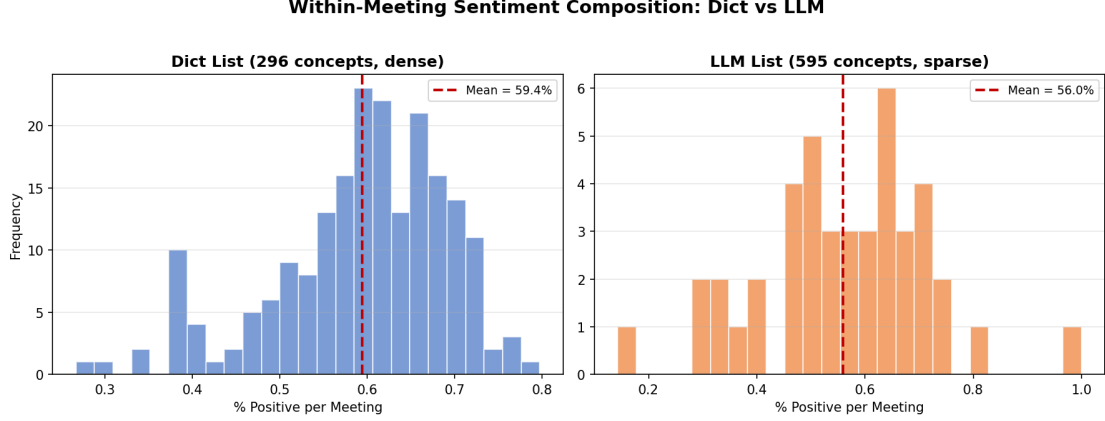


Figure 5: Per-meeting distribution of % positive scores. The LLM list (right) has higher variance, wider range, and a long left tail—consistent with Devesh’s intuition that it is less one-sided than the human list.

Baseline First-Stage Fit

We compare the incremental R^2 of each list’s PC1 in the first-stage regression:

Table 4: PC1 Incremental R^2 : Human vs. LLM List

Specification	R^2	ΔR^2
(A) Greenbook forecasts only	0.380	—
(B) + Dict PC1 (human 296)	0.458	+0.079
(C) + LLM PC1 (595)	0.411	+0.032
(D) + LLM all 595 cols	0.609	+0.229

The LLM PC1 adds less (+0.032) than the dict PC1 (+0.079), likely because its sparsity makes the first principal component noisier. However, using all 595 LLM columns directly (without PC1) produces a substantially larger R^2 gain (+0.229), though with $p = 595$ on $n = 204$, Ridge regularization is essential here.

Shuffle Test: PC1-Based

We replicate the exact same procedure as Q1 on the LLM matrix: (1) shuffle LLM sentiment columns within each meeting, (2) recompute PC1, (3) re-run Ridge with forecasts + shuffled PC1. 100 independent permutations. Results are compared against the same test on the human list:

Table 5: PC1 Shuffle Test: Human vs. LLM

List	Actual R^2	Shuffle Mean R^2	Shuffle Std	p -value
Human (296)	0.458	0.479	0.012	1.000
LLM (595)	0.411	0.407	0.016	0.713

Both lists show **no significant degradation** from shuffling. The human list actually sees a slight R^2 increase after shuffling (the shuffled PC1 captures slightly more FFR-predicting variation than the original). The LLM list shows a tiny drop (−0.004) that is well within one standard deviation. The p -values confirm neither effect is distinguishable from random permutation.

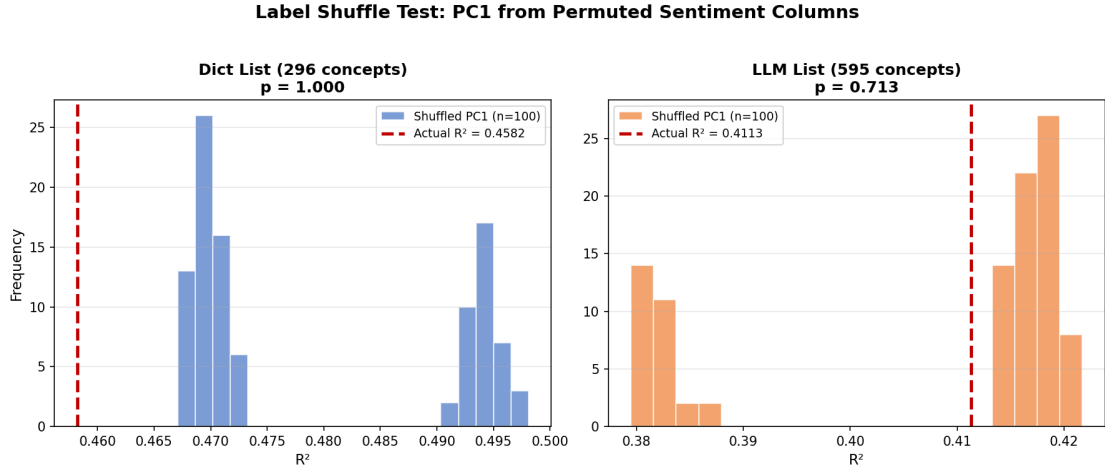


Figure 6: PC1 shuffle test distributions. The actual R^2 (red dashed line) lies within the shuffle distribution for both lists. For the human list, shuffling actually *increases* R^2 on average—the original label assignment is not optimal for predicting FFR changes.

Shuffle Test: Full Sentiment Columns

Using all sentiment columns directly (without PC1 reduction) tests whether concept-level heterogeneity provides fit beyond what the first component captures. We shuffle columns within each meeting and re-run Ridge directly on forecasts + shuffled sentiment columns:

Table 6: Full-Column Shuffle Test

List	Actual R^2	Shuffle Mean R^2	Shuffle Std	p -value
Human (296)	0.656	0.692	0.045	0.624
LLM (595)	0.609	0.626	0.0003	1.000

Both lists are robust to shuffling. The human list’s full-model shuffle produces a modest R^2 increase with meaningful variation across permutations (std = 0.045). The LLM list’s near-zero variation (std = 0.0003) reflects its extreme sparsity: with only ~ 5 non-zeros per row, shuffling these few scores across 595 columns produces nearly identical design matrices. This is a *limitation* of the LLM shuffle test—the sparsity that is a feature for concept identification becomes a constraint for permutation-based inference.

Shock Correlation

We compare the identified monetary policy shock from the original LLM sentiment matrix to the shock from each shuffled matrix:

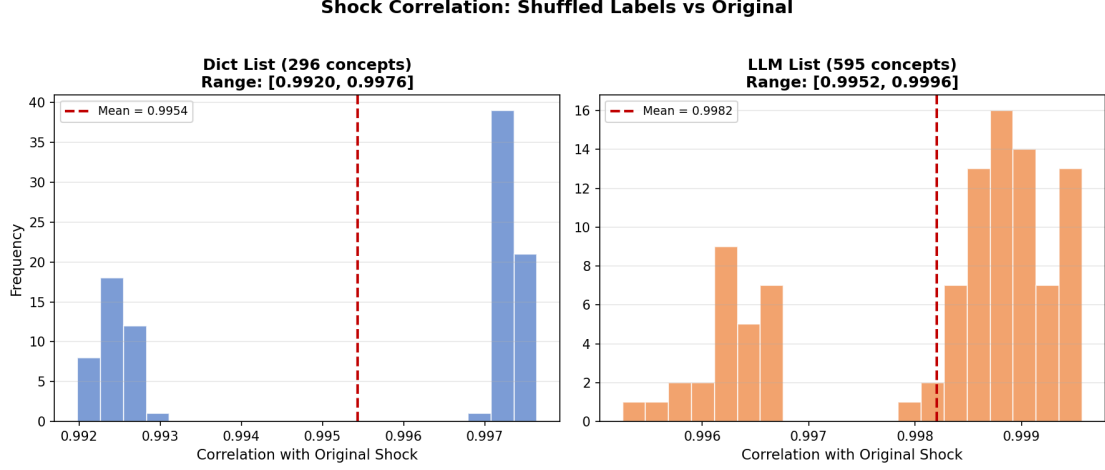


Figure 7: Shock correlation distributions: original vs. shuffled. Both lists produce near-perfect correlations, but the LLM list is even tighter ($r = 0.9982 \pm 0.0012$) because its extreme sparsity means shuffling barely alters the within-meeting score distribution.

Table 7: Shock Correlation: Original vs. Shuffled Labels

List	Mean r	Std	Range
Human (296)	0.9954	0.0024	[0.9920, 0.9976]
LLM (595)	0.9982	0.0012	[0.9952, 0.9996]

Interpretation

Devesh’s premise—that the LLM list’s more balanced composition makes the shuffle test more informative—is correct in principle. The LLM list has higher within-meeting score variance (std 0.156 vs. 0.099) and a wider range (14%–100% vs. 27%–80%). The issue is that the LLM list is simultaneously **99% sparse**, so the effective number of labels being permuted per meeting is only ~ 5 , not 595.

The result, however, is unambiguous: **neither list shows label-dependence**. The shock correlations are > 0.99 for both. The human list’s shuffle test even shows that randomized labels *improve* the first-stage R^2 —further evidence that the specific assignment of concept names to scores is irrelevant to Ridge’s ability to extract the document’s predictive signal.

This reinforces the conclusion from Q1–Q3: the number of measurement points matters, not which specific words anchor them.

Summary

1. **Q1: Shuffling does not change the Ridge fit.** Single-permutation shock correlations are tightly concentrated at $r = 0.947 \pm 0.005$. The fit depends on meeting-level document tone, which is invariant to column permutation. The 296 concepts matter through their *number* (providing 1,480 time-varying sampling points), not through their specific labels.
2. **Q2: Shuffled shocks produce the same IRFs.** The mathematical relationship $\tilde{\beta}_h = (\sigma_\varepsilon/\sigma_{\tilde{\varepsilon}}) \cdot \beta_h$ is confirmed empirically. AD(2024) shocks produce cleaner IRFs than Romer-Romer shocks, but the shuffled-label IRFs are essentially identical to the original.
3. **Q3: The model does not overfit.** The design matrix spans ~ 155 effective dimensions (not 3,224, not 1). Ridge regularization with $\lambda = 924$ successfully handles the p/n ratio.

Five diagnostic tests confirm this.

4. **Q4: The shuffle test holds across concept lists.** Replacing the human 296 concepts with an LLM-generated 595-concept list—which has more balanced within-meeting composition—does not change the result. Shock correlations remain > 0.99 . Both lists are invariant to label permutation. The LLM list’s extreme sparsity (99% zeros) limits the power of permutation-based inference but does not undermine the core finding: **labels are incidental; coverage is what matters.**

Code: `src/`. Results: `results/`. Presentation: `results/beamer_presentation.pdf`. All experiments use the same pipeline (sklearn RidgeCV, 10-fold CV, identical monthly conversion).